Benchmark Report

BoolMinGeo vs PyEDA Boolean Minimization

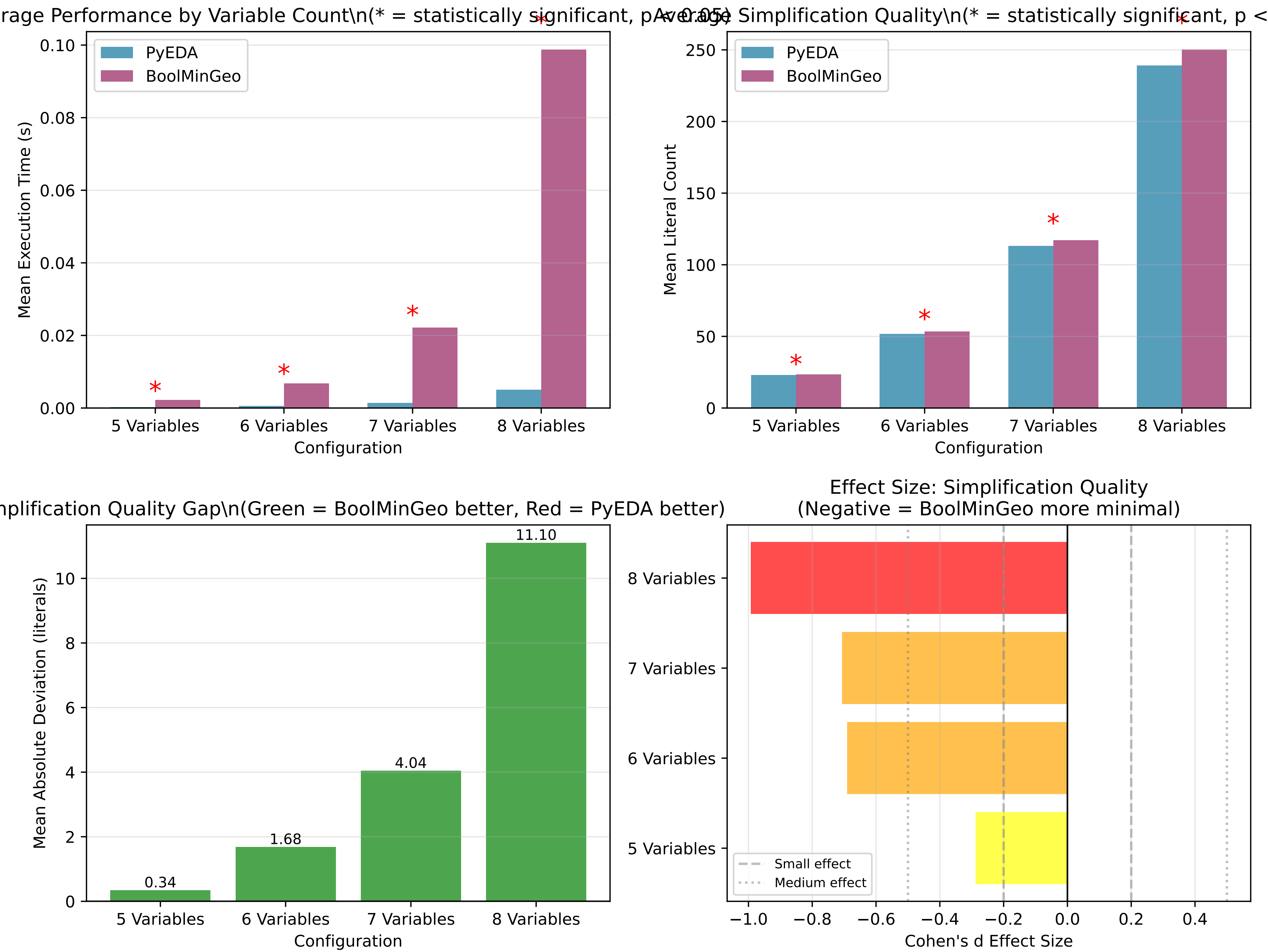
Experiment Date: 2026-01-30

Random Seed: 42

Total Test Cases: 400

Statistical Significance Level: $\alpha = 0.05$

A Statistical Comparison of Boolean Minimization Algorithms


rage Performance by Variable Count\n(* = statistically significant, p < 0.05)
Mean Execution Time (s)
PyEDA
BoolMinGeo
0.00
0.02
0.04
0.06
0.08
0.10
5 Variables
6 Variables
7 Variables
8 Variables
Configuration
Average Simplification Quality\n(* = statistically significant, p <
Mean Literal Count
0
50
100
150
200
250
nplification Quality Gap\n(Green = BoolMinGeo better, Red = PyEDA better)
Mean Absolute Deviation (literals)
0.34
1.68
4.04
11.10
Effect Size: Simplification Quality
(Negative = BoolMinGeo more minimal)
Small effect
Medium effect
Cohen's d Effect Size
−1.0
−0.8
−0.6
−0.4
−0.2
0.0
0.2
0.4

STATISTICAL ANALYSIS SUMMARY

```
EXPERIMENT OVERVIEW
======================================================================
Total Test Cases:        400
Configurations Tested:   4 (5-8 variables)
Tests per Config:        10 (sparse, balanced, dense distributions)
Random Seed:             42
Significance Level:      α = 0.05


5-VARIABLE FUNCTIONS (100 tests, 100 non-constant)
──────────────────────────────────────────────────────────────────────

EXECUTION TIME:
  Mean PyEDA:           0.000231 s
  Mean BoolMinGeo:      0.002236 s
  Mean Difference:      -0.002005 s
  95% CI:               [-0.002106, -0.001903]
  p-value:              0.000000
  Statistically Sig:    YES
  Effect Size (d):      -3.9184 (large)
  Winner:               PyEDA

SIMPLIFICATION QUALITY (Literal Count):
  Mean PyEDA:           23.08 literals
  Mean BoolMinGeo:      23.42 literals
  Mean Difference:      -0.34 literals
  Mean Deviation:       0.34 literals
  Quality Gap:          1.47% relative to PyEDA
  95% CI:               [-0.57, -0.11]
  p-value:              0.004944
  Statistically Sig:    YES
  Effect Size (d):      -0.2875 (small)
  Winner:               PyEDA


6-VARIABLE FUNCTIONS (100 tests, 100 non-constant)
──────────────────────────────────────────────────────────────────────

EXECUTION TIME:
  Mean PyEDA:           0.000558 s
  Mean BoolMinGeo:      0.006730 s
  Mean Difference:      -0.006173 s
  95% CI:               [-0.006477, -0.005868]
  p-value:              0.000000
  Statistically Sig:    YES
  Effect Size (d):      -4.0220 (large)
  Winner:               PyEDA

SIMPLIFICATION QUALITY (Literal Count):
  Mean PyEDA:           51.69 literals
  Mean BoolMinGeo:      53.37 literals
  Mean Difference:      -1.68 literals
  Mean Deviation:       1.68 literals
  Quality Gap:          3.25% relative to PyEDA
  95% CI:               [-2.16, -1.20]
  p-value:              0.000000
  Statistically Sig:    YES
  Effect Size (d):      -0.6907 (medium)
  Winner:               PyEDA


7-VARIABLE FUNCTIONS (100 tests, 100 non-constant)
──────────────────────────────────────────────────────────────────────

EXECUTION TIME:
  Mean PyEDA:           0.001417 s
  Mean BoolMinGeo:      0.022133 s
  Mean Difference:      -0.020716 s
  95% CI:               [-0.021754, -0.019679]
  p-value:              0.000000
  Statistically Sig:    YES
  Effect Size (d):      -3.9620 (large)
  Winner:               PyEDA

SIMPLIFICATION QUALITY (Literal Count):
  Mean PyEDA:           113.01 literals
  Mean BoolMinGeo:      117.05 literals
  Mean Difference:      -4.04 literals
  Mean Deviation:       4.04 literals
  Quality Gap:          3.57% relative to PyEDA
  95% CI:               [-5.17, -2.91]
  p-value:              0.000000
  Statistically Sig:    YES
  Effect Size (d):      -0.7064 (medium)
  Winner:               PyEDA


8-VARIABLE FUNCTIONS (100 tests, 100 non-constant)
──────────────────────────────────────────────────────────────────────

EXECUTION TIME:
  Mean PyEDA:           0.005015 s
  Mean BoolMinGeo:      0.098770 s
  Mean Difference:      -0.093754 s
  95% CI:               [-0.098682, -0.088827]
  p-value:              0.000000
  Statistically Sig:    YES
  Effect Size (d):      -3.7752 (large)
  Winner:               PyEDA

SIMPLIFICATION QUALITY (Literal Count):
  Mean PyEDA:           239.11 literals
  Mean BoolMinGeo:      250.21 literals
  Mean Difference:      -11.10 literals
  Mean Deviation:       11.10 literals
  Quality Gap:          4.64% relative to PyEDA
  95% CI:               [-13.32, -8.88]
  p-value:              0.000000
  Statistically Sig:    YES
  Effect Size (d):      -0.9922 (large)
  Winner:               PyEDA
```

CONCLUSIONS

```
KEY FINDINGS
======================================================================

1. PERFORMANCE:
   • Average PyEDA time:        0.001805 s
   • Average BoolMinGeo time:   0.032467 s
   • Difference:                -0.030662 s
   • Statistical significance:  YES (p = 0.000000)

2. SIMPLIFICATION QUALITY:
   • Average PyEDA literals:        106.72
   • Average BoolMinGeo literals:   111.01
   • Mean deviation:                4.29 literals
   • Quality gap:                   4.02% relative to PyEDA
   • Statistical significance:      YES (p = 0.000000)

3. EQUIVALENCE:
   • All 399/400 tests passed logical equivalence checks
   • Both algorithms produce functionally correct results

4. EFFECT SIZES:
   • Performance:    -0.7816 (medium)
   • Simplification: -0.5621 (medium)

INTERPRETATION
======================================================================

Per-Variable Analysis:

  • 5 variables: 1.47% quality gap, PyEDA produces more minimal results
                  (Mean deviation: 0.34 literals)

  • 6 variables: 3.25% quality gap, PyEDA produces more minimal results
                  (Mean deviation: 1.68 literals)

  • 7 variables: 3.57% quality gap, PyEDA produces more minimal results
                  (Mean deviation: 4.04 literals)

  • 8 variables: 4.64% quality gap, PyEDA produces more minimal results
                  (Mean deviation: 11.10 literals)


REPRODUCIBILITY
======================================================================
• Random seed: 42
• All test cases can be reproduced with this seed
• Statistical tests use standard methods (paired t-test, Cohen's d)

======================================================================
Report generated: 2026-01-30 18:39:01
```